

NADUN DE SILVA

Lead Software Engineer

 Auckland, New Zealand
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Summary

Lead Software Engineer with nearly 8 years of experience in cloud-native application development for Kubernetes and cloud platforms. Proven track record in architecting, developing, and deploying scalable applications, while ensuring site reliability in production environments. Skilled in defining technical direction and leading engineering teams to deliver high-impact solutions.

Experience

Lead Software Engineer

McCrae Tech

 April 2025 to Present

 Auckland, New Zealand

- Orchestrated the migration of Indexity's SRE frameworks and infrastructure from Orion Health into McCrae Tech, achieving a smooth transition without affecting end users.
 - Automated the customer request handling, incident handling, and on-call rotations for the Indexity data-planes, with comprehensive monitoring and alerting, significantly reduced response times.
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Senior Software Engineer

Orion Health

 November 2022 to April 2025

 Auckland, New Zealand

- Enhanced the phone number searches by implementing Apache Lucene-based indexing techniques, optimizing patient search performance across all customers.
 - Orchestrated comprehensive threat modeling and privacy assessments, leading to stronger safeguards, resulting in zero major vulnerabilities during customer-initiated penetration tests.
 - Developed the Terraform code for deploying self-hosted GitLab runners on Google Cloud, saving more than 50% of the development costs.
 - Directed the successful deployment of Indexity data-planes using AWS infrastructure; conducted thorough disaster recovery planning, which reduced potential downtime risks from unforeseen incidents by at least 40%.
 - Championed an end-to-end deployment strategy for Indexity within a deployment framework on GitLab and AWS, resulting in a faster rollout time that decreased development cycles by one day per development cycle.
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Associate Technical Lead

WSO2

 June 2021 to November 2022

 Colombo, Sri Lanka

- Secured the Sustained Outstanding Contribution Award three consecutive years, an honor given exclusively to only the top 5% of all company staff members who demonstrated exceptional technical leadership and innovative contributions throughout tenure.
- Led a Senior Software Engineer in developing the minimum viable features for the resource scheduling of the Choreo online editor within 1.5 months, using Kubernetes and Go.
- Eliminated bottlenecks, reducing the startup time of the Kubernetes resources of the Choreo Editors by 80% and significantly enhancing user experience.
- Reduced the MSSQL database utilization by 60% using a Redis cache and randomization of cache expiry times, increasing the number of users the system can handle.
- Led the product team, completing 95% of the team targets and sprint milestones on time by prioritizing tasks and fostering a good working environment.

Senior Software Engineer

WSO2

📅 July 2019 to June 2021

📍 Colombo, Sri Lanka

- Spearheaded the implementation of the foundation for Choreo observability within 3 months with a team of 2 other engineers, creating the backbone of Choreo observability.
 - Decreased the cost by 90% for the company by architecting the Choreo observability storages, including data archival into a Data Lake for Machine Learning (ML) use cases.
 - Improved debugging experience for users by revamping the observability instrumentation at the Ballerina compiler level, within 1 month, to map the observability data to the source code.
 - Minimized the number of bugs in the Choreo Observability area by implementing proper code reviewing, testing, and deployment practices in a team of 6 engineers.
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Software Engineer

WSO2

📅 January 2018 to July 2019

📍 Colombo, Sri Lanka

- Delivered the Cellery observability basic features within 2 months for observing microservice composites using Kubernetes, Istio, OpenTracing, and Envoy.
- Headed the implementation of Cellery developer tools using VS Code Language Server Extensions and visualizations of Cells using D3.js.
- Developed Cellery Hub backed by a Docker Registry as the storage and implemented the authentication of the CLI and portal using OpenID Connect (OIDC) within 1 month.
- Implemented the observability aspects of the WSO2 Serverless Platform using Prometheus and Jaeger on top of Kubernetes and OpenWhisk.

Skills

- **Programming languages** — Java, Go, Scala, Python, JavaScript, TypeScript
- **Cloud Platforms** — AWS, GCP, Azure
- **Containerization** — Kubernetes, Azure Kubernetes Service, AWS ECS, Docker, Kustomize, Helm
- **Frameworks and tools** — Terraform, Kubernetes Operator Framework, React, Ansible, ExpressJS
- **Data Storage** — Time-series Databases (InfluxDB, Azure Data Explorer), Graph Databases (OrientDB), Data Lakes (Azure Data Lake), Relational Databases (MySQL, Microsoft SQL Server), Redis
- **CI/CD Tools and Practices** — GitOps, GitHub Actions, GitLab
- **Observability Tools and Platforms** — OpenTelemetry, DataDog, Prometheus, Jaeger, CloudWatch
- **API Protocols** — REST, GraphQL, gRPC
- **Software Development Methodologies** — Agile, DevOps

Education

B.Sc. (Hons.) in Engineering (Computer Science and Engineering)

University of Moratuwa

📅 January 2014 to December 2017

📍 Colombo, Sri Lanka

- Attained a GPA of 3.85 out of 4.20, obtaining a First Class.
- Placements in Dean's List in 6 out of 8 semesters at the University of Moratuwa.
- Awarded Global Finalist (Galactic Impact) in the NASA Space Apps Challenge 2017.
- Completed Google Summer of Code 2017.

Certifications

Certified Kubernetes Administrator

The Linux Foundation

📅 December 2020

Certified Kubernetes Application Developer

The Linux Foundation

📅 January 2020